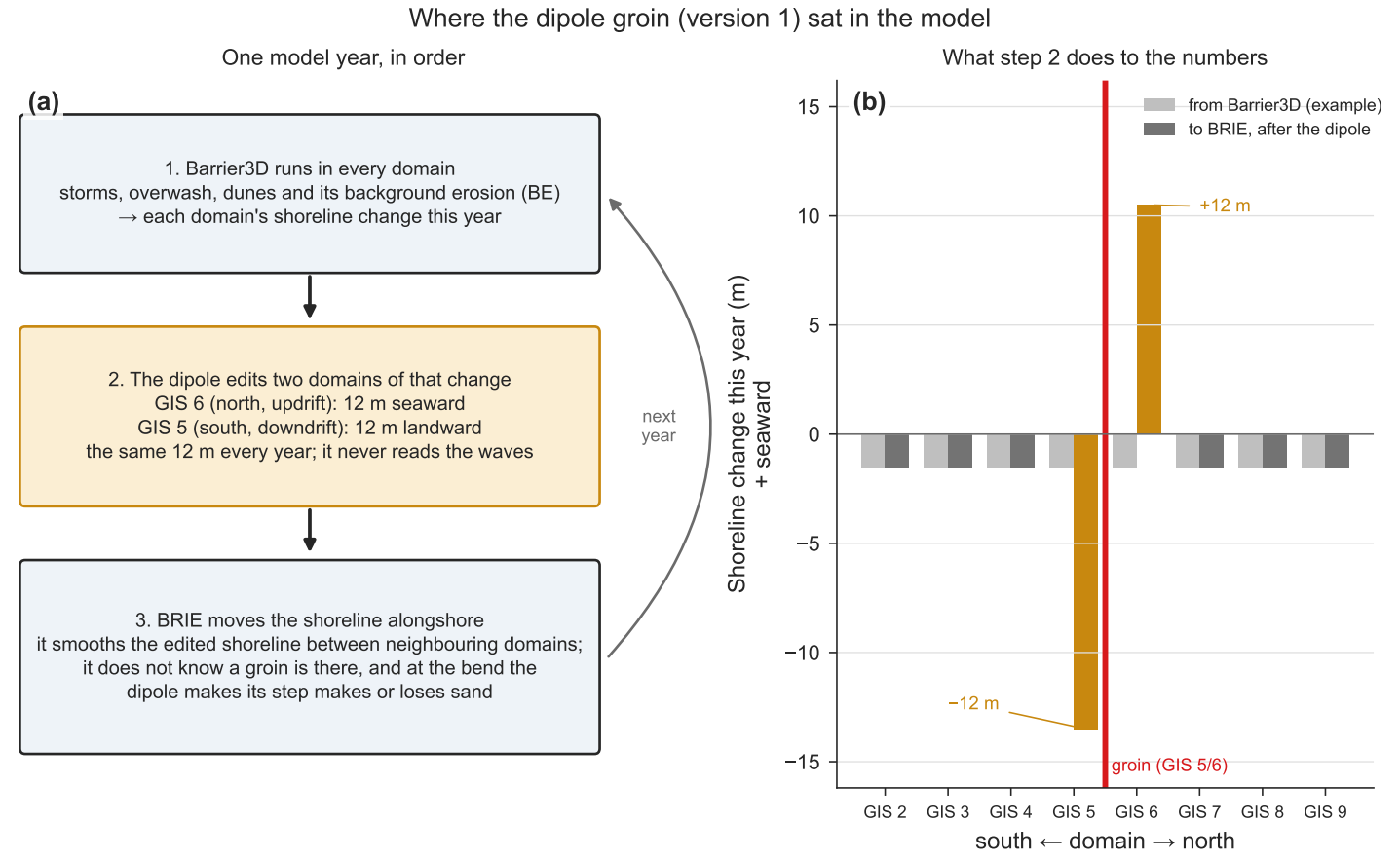
Where the dipole groin (version 1) sat in the model
One model year, in order
What step 2 does to the numbers
(a)
1. Barrier3D runs in every domain
storms, overwash, dunes and its background erosion (BE)
→ each domain's shoreline change this year
2. The dipole edits two domains of that change
GIS 6 (north, updrift): 12 m seaward
GIS 5 (south, downdrift): 12 m landward
the same 12 m every year; it never reads the waves
3. BRIE moves the shoreline alongshore
it smooths the edited shoreline between neighbouring domains;
it does not know a groin is there, and at the bend the
dipole makes its step makes or loses sand
next
year
(b)
from Barrier3D (example)
to BRIE, after the dipole
+12 m
−12 m
groin (GIS 5/6)
Shoreline change this year (m)
+ seaward
15
10
5
0
−5
−10
−15
GIS 2
GIS 3
GIS 4
GIS 5
GIS 6
GIS 7
GIS 8
GIS 9
south ← domain → north